Q-MAVLink: A qSIM-Based Entity Authentication and Post-Quantum Key Exchange Protocol *

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Abstract

MAVLink v2 provides an optional Sign field but lacks sufficient confidentiality, entity authentication, and key exchange mechanisms. This paper proposes Q-MAVLink, which combines qSIM-based PSK mutual authentication with ML-KEM-512-based key exchange. The proposed protocol establishes a shared secret between the UAV and GCS, enabling integration with existing MAVLink payload encryption schemes to provide confidentiality, entity authentication, authenticated key establishment, and post-quantum security.

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